

Lightweight Neural Network for Real-Time Mouth Movement Detection

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Abstract

This project proposes the development of a lightweight feedforward neural network for real-time binary classification of mouth movement states. Using backpropagation-based training on facial landmark features, the system will distinguish between moving and non-moving mouths with high accuracy and minimal computational overhead. The application targets enhancement of speech-to-text systems by identifying active speakers in multi-person environments. The neural network architecture will be optimized for real-time video processing (30+ fps) on standard hardware, making it suitable for deployment in video conferencing, automated transcription, and accessibility applications.

I. INTRODUCTION

Identifying which person is speaking in multi-person environments remains a fundamental challenge for speech-to-text (STT) systems. While audio-based speaker diarization techniques exist, they struggle with overlapping speech, similar voice characteristics, and background noise. Visual information from mouth movements provides a robust complementary signal that is largely invariant to acoustic conditions.

This project focuses specifically on the mouth movement detection component, a binary classification problem that determines whether a person's mouth is currently moving (potentially speaking) or stationary. By framing this as a supervised learning task, we can leverage neural networks trained via backpropagation to learn discriminative patterns from facial features. The lightweight design ensures real-time performance suitable for live video applications.

The key innovation lies in developing a compact neural network architecture that balances classification accuracy with computational efficiency. Unlike deep learning approaches requiring extensive computational resources, our feedforward network with a single hidden layer aims to achieve >90% accuracy while maintaining inference times under 10ms per frame. This makes the system practical for integration into existing video conferencing and transcription workflows as a preprocessing step for future STT enhancement.

II. PROJECT OBJECTIVES AND TASK DESCRIPTION

The primary objective is to develop a lightweight neural network that accurately performs binary classification of mouth states in real-time video. Specific objectives include:

1. Design a feedforward neural network with backpropagation training that classifies mouth states as "moving" (1) or "not moving" (0)
2. Extract discriminative facial features from video frames that correlate with mouth movement
3. Achieve classification accuracy of at least 90% on held-out test data
4. Optimize the network architecture for real-time performance (≥ 30 fps inference) on standard CPU hardware
5. Evaluate robustness across varied lighting conditions, head poses, and speaker characteristics
6. Minimize false positives from non-speech mouth movements (e.g., facial expressions, eating)

This project addresses a supervised binary classification task. Given a video frame containing a face, the system must output a binary prediction indicating whether the mouth is moving. The scope is limited to the mouth detection component only, integration with STT systems is considered a future extension beyond this phase. The neural network will be trained on labeled video data where ground truth labels are derived from audio activity analysis and manual verification.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of four main processing stages in a pipeline configuration:

A. Video Input and Preprocessing

Captures video frames from camera or file at 30+ fps. Performs color space conversion (RGB to grayscale for efficiency) and frame buffering for temporal analysis.

B. Face Detection and Landmark Localization

Detects faces in each frame using efficient algorithms (Haar cascades or MTCNN) and localizes 68 facial landmarks using regression tree ensembles (dlib implementation). Approximately 20 landmarks delineate the

mouth region, providing precise geometric coordinates of inner and outer lip contours. These landmarks enable calculation of mouth-specific features while implicitly handling the mouth region extraction. The landmark tracking across frames maintains consistency for temporal feature computation.

C. Feature Extraction Module

Computes discriminative features from mouth landmarks and pixel intensities:

- Mouth Aspect Ratio (MAR): Ratio of vertical to horizontal mouth opening
- Inter-landmark distances: Euclidean distances between key mouth points
- Temporal derivatives: Frame-to-frame changes in landmark positions (optical flow)
- Pixel intensity statistics: Mean and variance within mouth bounding box
- Edge responses: Gradient magnitudes indicating mouth contour definition

The feature vector (20-30 dimensions) is normalized to zero mean and unit variance for neural network input.

D. Neural Network Classification

A compact feedforward neural network trained via backpropagation performs binary classification:

Architecture:

- Input layer: 20-30 neurons (feature vector dimensionality)
- Hidden layer: 10-15 neurons with sigmoid activation function
- Output layer: 1 neuron with sigmoid activation (outputs probability in $[0,1]$)
- Training: Backpropagation algorithm with binary cross-entropy loss, gradient descent optimization, learning rate 0.01-0.1 (tuned via validation), batch size 32-64. Early stopping based on validation loss prevents overfitting.
- Classification threshold: Outputs ≥ 0.5 classified as "moving" (1), otherwise "not moving" (0). Confidence scores enable downstream filtering.

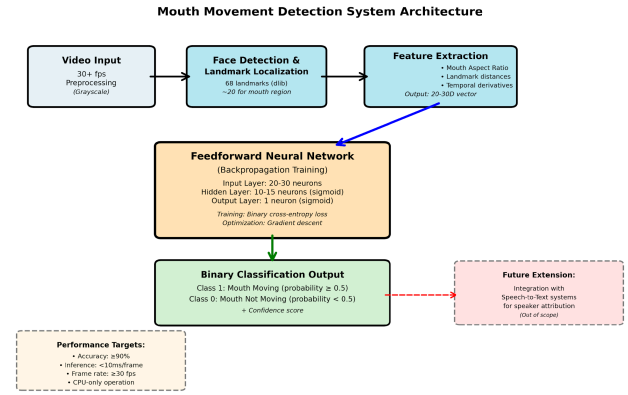


Fig. 1. System architecture showing the four-stage pipeline for mouth movement detection. The neural network performs binary classification with real-time performance targets.

IV. NEURAL NETWORK DESIGN RATIONALE

The choice of a feedforward neural network with backpropagation training is motivated by several key factors relevant to this binary classification task:

A. Advantages for Mouth Movement Detection

Neural networks excel at learning non-linear decision boundaries from high-dimensional feature spaces. Mouth movement involves complex patterns combining geometric (landmark positions), temporal (frame-to-frame changes), and intensity-based features. The hidden layer with sigmoid activation functions enables the network to learn intermediate representations that capture these non-linear relationships.

The backpropagation algorithm provides an efficient mechanism for training multilayer networks through gradient-based optimization. By computing gradients via the chain rule and propagating errors backward through the network, backpropagation adjusts weights to minimize classification error. This iterative refinement converges to locally optimal solutions suitable for binary classification tasks.

The compact architecture (single hidden layer with 10-15 neurons) ensures computational efficiency while maintaining sufficient representational capacity. This design contrasts with deep learning approaches requiring extensive computational resources, making our network suitable for CPU-only inference at video frame rates.

B. Comparison to Alternative Approaches

While decision trees and k-nearest neighbors offer interpretability and simplicity, neural networks provide superior performance for this application. Decision trees risk overfitting to specific landmark configurations and may struggle with continuous-valued temporal features. k-NN

requires storing the entire training set and computing distances at inference time, creating computational bottlenecks for real-time video processing. Neural networks naturally handle continuous feature spaces, learn hierarchical representations, and achieve fast inference through simple matrix multiplications and activation functions. The trained network requires minimal memory (only weight matrices), enabling efficient deployment.

V. DATASET AND BENCHMARK

The project will utilize multiple data sources for comprehensive training and evaluation:

A. LRS2 Dataset (Primary Source)

The Lip Reading Sentences 2 (LRS2) dataset from Oxford University provides large-scale natural speech video data. Containing over 140,000 utterances from BBC programs, LRS2 offers diverse speakers, head poses, and lighting conditions representative of real-world scenarios. We will extract frames and generate binary labels based on audio activity: frames with speech energy above threshold are labeled "moving" (1), silent frames labeled "not moving" (0). This automatic labeling will be validated through manual inspection of sample subsets.

B. Custom Collected Dataset (Validation Source)

To ensure robust evaluation, we will collect controlled video data:

- 3-5 participants, 5 minutes speaking + 3 minutes silence each
- Varied conditions: bright/dim lighting, different head angles ($\pm 30^\circ$), multiple distances
- Inclusion of non-speech mouth movements (yawning, chewing) to test false positive rates
- Manual frame-level annotation for ground truth accuracy

C. Data Split and Class Balance

Total dataset: ~1,500 labeled frames

- Training set: 70% (1,050 frames)
- Validation set: 15% (225 frames)
- Test set: 15% (225 frames)

Class distribution will be balanced at 50% moving / 50% not moving through stratified sampling. This prevents bias toward the majority class and ensures the network learns discriminative features rather than dataset statistics.

VI. IMPLEMENTATION DETAILS

Programming Language: Python 3.8+

Core Libraries:

- NumPy: Matrix operations and numerical computations for backpropagation
- OpenCV: Video I/O, face detection, image preprocessing
- dlib: 68-point facial landmark detection
- scikit-learn: Feature normalization, train-test splitting, evaluation metrics

The neural network will be implemented from scratch in NumPy to demonstrate understanding of backpropagation mechanics. Weight matrices will be initialized using He initialization. The forward pass computes activations through matrix multiplications and sigmoid functions. The backward pass implements the chain rule to compute gradients and update weights via gradient descent.

Development Environment: Jupyter Notebook for experimentation, Python scripts for production pipeline. Version control via Git.

VII. EVALUATION PLAN

The system will undergo rigorous quantitative and qualitative evaluation:

A. Quantitative Performance Metrics

1. Classification Accuracy: Percentage of correctly classified frames on test set (target: $\geq 90\%$)
2. Precision: True positives / (True positives + False positives) for "moving" class
3. Recall: True positives / (True positives + False negatives) for "moving" class
4. F1-Score: Harmonic mean of precision and recall, balancing both metrics
5. Confusion Matrix: Detailed breakdown of classification errors
6. ROC Curve and AUC: Evaluate classification performance across threshold values
7. Inference Time: Average milliseconds per frame (target: $< 10\text{ms}$ on CPU)
8. Throughput: Frames processed per second (target: $\geq 30\text{ fps}$)

B. Robustness Analysis

Systematic evaluation across challenging conditions:

Lighting variation: Performance in bright, dim, and backlit scenarios

- Head pose: Accuracy at $\pm 15^\circ$, $\pm 30^\circ$ rotations from frontal view
- Speaking style: Fast speech, slow speech, whispering
- False positive analysis: Classification of non-speech movements (yawning, chewing, facial expressions)

- Occlusions: Partial face visibility (hands near mouth, glasses)

C. Success Criteria

The project will be considered successful if:

1. Test set accuracy exceeds 90%
2. F1-score exceeds 0.85, demonstrating balanced precision and recall
3. Inference time remains under 10ms per frame on standard CPU
4. System maintains >85% accuracy under varied lighting and pose conditions
5. False positive rate for non-speech movements stays below 15%

VIII. REFERENCES

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IX. ANTICIPATED CHALLENGES AND MITIGATION STRATEGIES

A. Technical Challenges

1. Balancing accuracy with computational efficiency for real-time performance
2. Handling illumination variations that affect landmark detection quality
3. Distinguishing speech movements from other mouth motions (eating, expressions)
4. Maintaining accuracy across diverse head poses and speaker characteristics
5. Preventing overfitting given limited training data

B. Mitigation Approaches

- Architectural optimization: Limit network depth and width based on validation performance
- Data augmentation: Apply brightness/contrast adjustments, geometric transformations during training

- Temporal smoothing: Use sliding window averaging to reduce frame-to-frame noise
- Regularization: Implement L2 weight decay and early stopping to prevent overfitting
- Feature engineering: Include temporal derivatives to emphasize speech-specific dynamics
- Robust landmark detection: Use MTCNN or similar methods with built-in pose invariance

C. Future Extensions (Beyond Current Scope)

While not part of this phase, potential extensions include:

- Integration with speech-to-text systems for speaker attribution in multi-person conversations
- Multi-class classification: Distinguish speech intensity levels (whisper, normal, loud)
- Recurrent architectures (LSTM/GRU) for improved temporal modeling
- Edge deployment: Optimization for mobile and embedded devices
- Active learning: Iterative model improvement with user feedback

X. CONCLUSION

This project proposes a focused approach to mouth movement detection using a lightweight feedforward neural network trained via backpropagation. By concentrating solely on the binary classification task, determining whether a mouth is moving or stationary, we can develop a robust, computationally efficient system suitable for real-time video processing. The compact architecture (single hidden layer with 10-15 neurons) balances representational capacity with inference speed, targeting >90% accuracy while maintaining <10ms per-frame latency on CPU hardware.

The proposed system leverages established techniques from computer vision (facial landmark detection) and machine learning (backpropagation-based neural networks from ELE 588 Lecture 3) to create a practical solution. Through systematic evaluation on diverse datasets (LRS2 and custom-collected data) across varied conditions, we will demonstrate the system's robustness and real-world applicability.

While integration with speech-to-text systems represents a natural future direction, this project establishes the foundational mouth detection component. Success in this phase, achieving accurate, real-time binary classification—will validate the approach and enable subsequent work on multi-modal speaker identification. The lightweight design philosophy ensures the system remains accessible for deployment across diverse applications, from video conferencing to accessibility tools, without requiring specialized hardware.